2520090025

Section-11

D Nikhil Krishna

```
nikhilkrishna@NIKHIL:~$ sudo apt update
sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
169 packages can be upgraded. Run 'apt list --upgradable' to see them.
tshark is already the newest version (4.6.4-1).
iperf3 is already the newest version (3.20-2).
mtr-tiny is already the newest version (0.95-1.1ubuntu2).
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
wireshark is already the newest version (4.6.4-1).
netcat-openbsd is already the newest version (1.234-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 169
```

```
nikhilkrishna@NIKHIL:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
[sudo: authenticate] Password:
nikhilkrishna@NIKHIL:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
nikhilkrishna@NIKHIL:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
nikhilkrishna@NIKHIL:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
nikhilkrishna@NIKHIL:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
nikhilkrishna@NIKHIL:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
nikhilkrishna@NIKHIL:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
nikhilkrishna@NIKHIL:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.027 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.094 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.040 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2028ms
rtt min/avg/max/mdev = 0.027/0.053/0.094/0.029 ms
nikhilkrishna@NIKHIL:~$
```

```
nikhilkrishna@NIKHIL:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
9 Hello from host1!
71 ^C
nikhilkrishna@NIKHIL:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
nikhilkrishna@NIKHIL:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
     21 63
     24 64
     21 63
     24 64
     24 63
     21 64
nikhilkrishna@NIKHIL:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e data.text
2>/dev/null
```

```
nikhilkrishna@NIKHIL:~$ sudo apt install -y tshark iproute2 netcat-openbsd wireshark
tshark is already the newest version (4.6.4-1).
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
netcat-openbsd is already the newest version (1.234-1).
wireshark is already the newest version (4.6.4-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 176
nikhilkrishna@NIKHIL:~$ sudo ip netns list
host2 (id: 2)
router (id: 1)
host1 (id: 0)
nikhilkrishna@NIKHIL:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
RTNETLINK answers: File exists
RTNETLINK answers: File exists
nikhilkrishna@NIKHIL:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
nikhilkrishna@NIKHIL:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
Error: ipv4: Address already assigned.
nikhilkrishna@NIKHIL:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
```

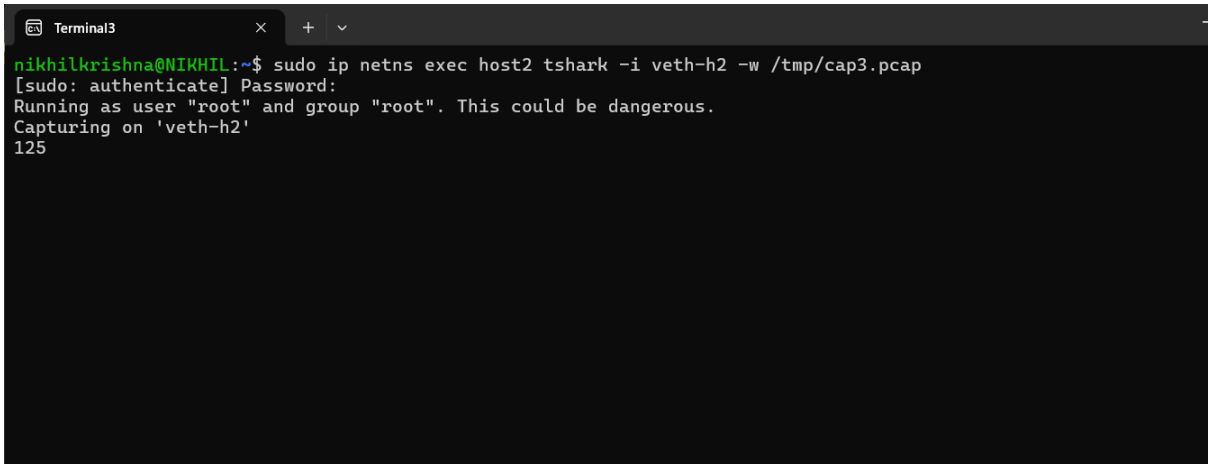
```
nikhilkrishna@NIKHIL:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
Error: ipv4: Address already assigned.
nikhilkrishna@NIKHIL:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
RTNETLINK answers: File exists
RTNETLINK answers: File exists
nikhilkrishna@NIKHIL:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.043 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.107 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.055 ms

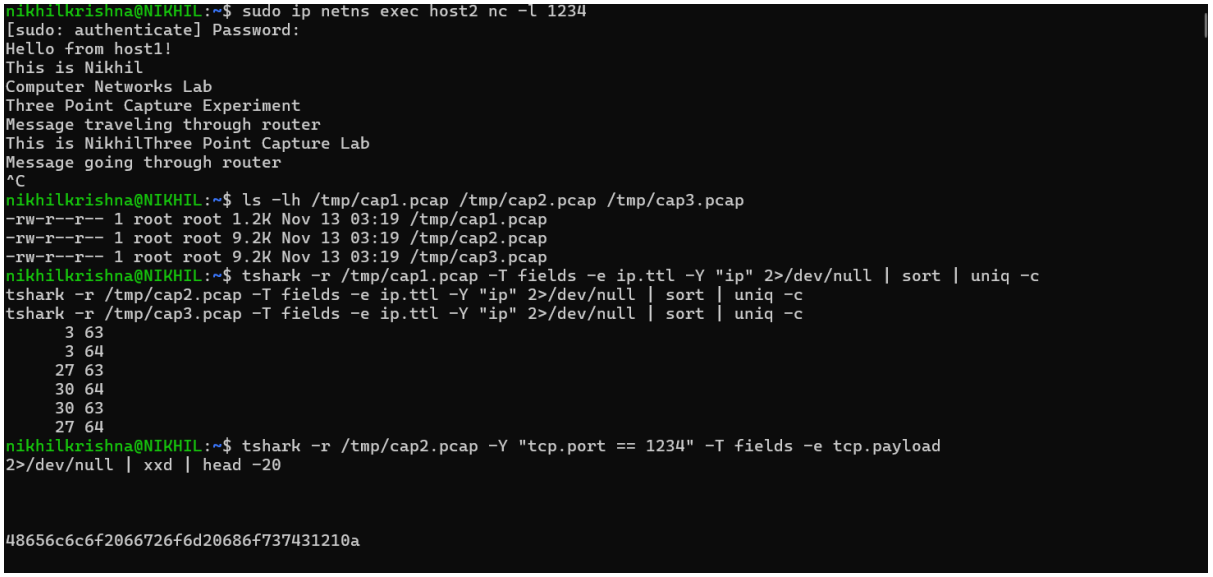
--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2054ms
rtt min/avg/max/mdev = 0.043/0.068/0.107/0.027 ms
nikhilkrishna@NIKHIL:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss
1%
Command line is not complete. Try option "help"
1%: command not found
```

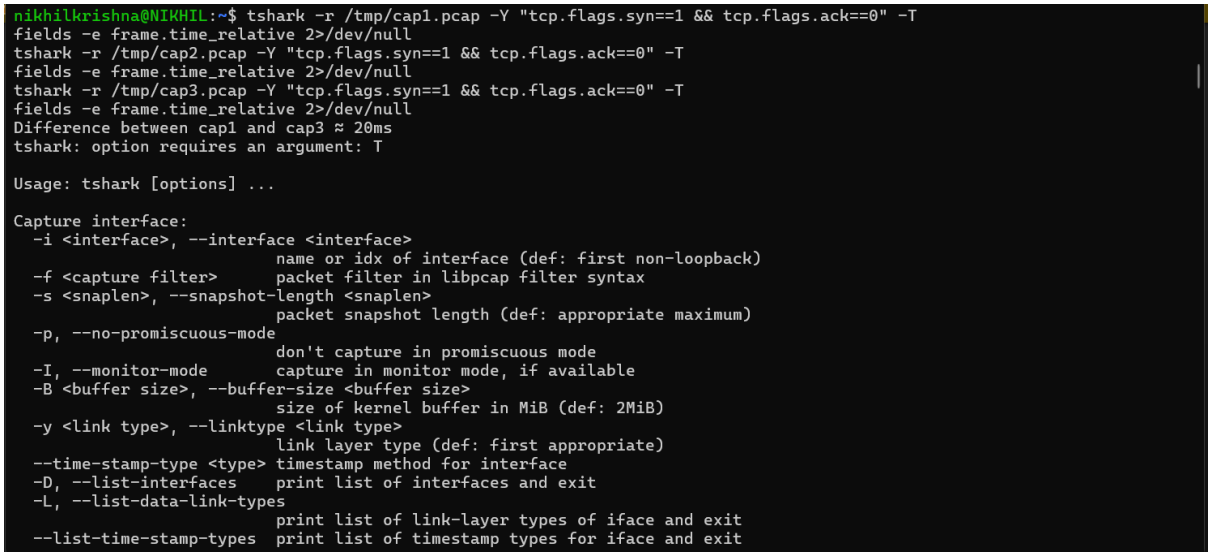
```
nikhilkrishna@NIKHIL:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss
1%
Command line is not complete. Try option "help"
1%: command not found
nikhilkrishna@NIKHIL:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc noqueue 0: root refcnt 2
nikhilkrishna@NIKHIL:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
62
```

Terminal2

```
nikhilkrishna@NIKHIL:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
145
```


Terminal3
nikhilkrishna@NIKHIL:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
125


nikhilkrishna@NIKHIL:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
Hello from host1!
This is Nikhil
Computer Networks Lab
Three Point Capture Experiment
Message traveling through router
This is NikhilThree Point Capture Lab
Message going through router
^C
nikhilkrishna@NIKHIL:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 1.2K Nov 13 03:19 /tmp/cap1.pcap
-rw-r--r-- 1 root root 9.2K Nov 13 03:19 /tmp/cap2.pcap
-rw-r--r-- 1 root root 9.2K Nov 13 03:19 /tmp/cap3.pcap
nikhilkrishna@NIKHIL:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
3 63
3 64
27 63
30 64
30 63
27 64
nikhilkrishna@NIKHIL:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e tcp.payload
2>/dev/null | xxd | head -20
48656c6c6f2066726f6d20686f737431210a


nikhilkrishna@NIKHIL:~$ tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T
fields -e frame.time_relative 2>/dev/null
tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T
fields -e frame.time_relative 2>/dev/null
tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T
fields -e frame.time_relative 2>/dev/null
Difference between cap1 and cap3 ≈ 20ms
tshark: option requires an argument: T
Usage: tshark [options] ...
Capture interface:
-i <interface>, --interface <interface>
name or idx of interface (def: first non-loopback)
-f <capture filter> packet filter in libpcap filter syntax
-s <snaplen>, --snapshot-length <snaplen>
packet snapshot length (def: appropriate maximum)
-p, --no-promiscuous-mode
don't capture in promiscuous mode
-I, --monitor-mode capture in monitor mode, if available
-B <buffer size>, --buffer-size <buffer size>
size of kernel buffer in MiB (def: 2MiB)
-y <link type>, --linktype <link type>
link layer type (def: first appropriate)
--time-stamp-type <type> timestamp method for interface
-D, --list-interfaces print list of interfaces and exit
-L, --list-data-link-types
print list of link-layer types of iface and exit
--list-time-stamp-types print list of timestamp types for iface and exit

```
nikhilkrishna@NIKHIL:~$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
cap3 shows highest count - loss happened between router and host2.
0
0
0
Command 'cap3' not found, did you mean:
  command 'cap' from deb capistrano (3.20.0-1)
Try: sudo apt install <deb name>
nikhilkrishna@NIKHIL:~$ sudo ip netns exec host1 mtr --report 10.0.2.2
Start: 2026-11-13T03:20:36+0000
^Z
[1]+  Stopped                 sudo ip netns exec host1 mtr --report 10.0.2.2
nikhilkrishna@NIKHIL:~$ wireshark /tmp/cap2.pcap &
[2] 2935
```

Terminal5

```
nikhilkrishna@NIKHIL:~$ sudo ip netns exec host1 nc 10.0.2.2 1234
[sudo: authenticate] Password:
Hello from host1!
This is Nikhil
Computer Networks Lab
Three Point Capture Experiment
Message traveling through router
This is NikhilThree Point Capture Lab
Message going through router
```

